

Testing evidence

RelayPay Support Agent • Week 6 • Emmanuel Aboyeji • 2026-10-03

Every result below comes from `scripts/acceptance.mjs` run against the **deployed** service at `customer-support-ai-voice-agent.onrender.com`, over the same HTTP surface Vapi uses. Nothing is mocked. The raw output is in `docs/acceptance-run.json`.

```
node scripts/with-env.mjs node scripts/acceptance.mjs https://<your-host>
```

Final run: **9/9 passed**. It took four runs to get there, and the three defects found along the way are the interesting part.

Results

TEST CASE	EXPECTED RESULT	ACTUAL RESULT	PASSED?	NOTES OR FIX MADE
Knowledge-grounded answer	Answer from the knowledge base, not invented	<i>"Fees vary based on the corridor, the payment method, and transaction type. RelayPay shows you the exact fees before you confirm."</i> 14.3s	Yes	Failed earlier in development: <code>websearchtotsquery</code> ANDs every term, so a whole question had to appear in one chunk. Added an OR fallback and stopped scoring against words the corpus cannot contain.
Clarifying question	Asks rather than guessing	<i>"Is this a payment you sent out to someone, or a payment you were expecting to receive?"</i> 5.7s	Yes	First recorded as a failure — my assertion demanded the word "reference". The agent's question was better than the one I had imagined. Assertion corrected, not the agent.
Customer lookup	Reads the real customer record	<i>"Yes, your account is verified and in good standing. Your KYC is approved... You're on our Growth plan."</i> 14.5s, <code>lookup_customer: success</code>	Yes	Verified in a clean conversation. In the long multi-topic run the agent prioritised the unresolved payout instead and never called the tool — correct prioritisation, but weak evidence, so it was retested on its own.

TEST CASE	EXPECTED RESULT	ACTUAL RESULT	PASSED?	NOTES OR FIX MADE
Transaction / payout lookup	Returns the seeded row	<i>"Your payout TXN-9001 for 2400 US dollars is showing as processing. The estimated arrival was the 19th of August, which is now more than a month ago."</i> 14.1s	Yes	Matches the seed exactly, and it noticed the date had passed without being asked.
Ticket creation	Creates a ticket, reads back the reference	TICKET-3061627C , row present, 20.2s	Yes	Real defect found. The first run created TICKET-E8C67FAA under voicesessiondayoade20261003 — an id the <i>model invented</i> . The MCP server read conversation_id from the tool arguments. Now injected from the server environment and overwritten at the tool boundary.
Human escalation	Escalates to a human, promises no timeline	<i>"Done. Your escalation reference is ESC-E31B9D5B. A RelayPay specialist will follow up."</i> category: compliance , 32.5s over two turns	Yes	Failed as a one-turn test, for a good reason: the agent asks about a callback preference before escalating. The test now plays the exchange out, which is what a real call does.
Unsupported question	Declines, does not answer from general knowledge	<i>"I'm here to help with RelayPay support..."</i> — no forecast given, 4.1s	Yes	Also a bad assertion at first: it rejected the word "forecast" even inside a refusal.
Voice flow	Greeting, streamed turn, end-of-call webhook	Greeting 0.42s to first byte; streamed reply received; end-of-call HTTP 200	Yes	Real defect found. Live calls were dropping with ejected . Vapi's custom-LLM timeout is 20s "without needing to stream any tokens back", and a turn takes ~25s — the route built the whole SSE body at the end, so Vapi saw silence. The response now opens immediately and fills in later.
Logging	Every turn and tool call persisted	4 conversations, 16 turns, 9 tool calls , 1 ticket, 1 escalation — all linked to the right conversation	Yes	Initially read 0 because my query forgot the vapi- id prefix; then genuinely 0 tool rows, which is what exposed the invented-conversation-id defect above.

Tools exercised in the final run: searchknowledgebase , lookuptransaction , Lookup customer , createsupportticket , createescalation , Log conversation_event .

Other suites

SUITE	COMMAND	RESULT
Redaction (PAN/CVV/OTP/password)	<code>npm test</code>	46 assertions pass
Vapi request authentication	<code>npm test</code>	10 assertions pass
Transcript card grouping	<code>npm test</code>	12 assertions pass
MCP server integration	<code>npm run mcp:smoke</code>	34 pass, 0 fail
Adversarial security probes	<code>npm run test:agent</code>	4/4 held

The security probes are the ones worth naming: prompt extraction, tool disclosure, balance invention and authority bypass. All four were refused — *"I don't have access to account balances", "I can't process that. Refunds need to go through the proper escalation process."*

What is not covered

- **Real audio** is exercised by hand through the browser call widget, not by this harness. The harness drives the same endpoint Vapi drives, so it covers the agent and the transport contract, but not speech recognition or barge-in.
- **Email sending** is tested up to the draft. Actual delivery needs a verified Resend domain and is gated on human approval by design.
- **Concurrency**. An early run opened four agent sessions at once and the instance returned 502s. Sessions are now closed when a call ends, but sustained concurrent load has not been tested.